

LLM Reasoning Coherence and Robustness: An Argumentation Theory Approach

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Abstract

This document contains the results of a research experiment using argumentation theory to analyze Large Language Model reasoning coherence and quantify the robustness of test-time compute. We select a frontier open-weight model, LLaMA-3.3-70B-Versatile, and generate adversarial perturbations on exercises sampled from the well known StrategyQA and GSMK8 datasets. We then task the model with both the perturbed and unperturbed exercises and use the "LLM-as-a-Judge" paradigm to evaluate its reasoning traces against the Toulmin model of argumentation. By assessing whether each reasoning step provides sufficient grounds and a valid warrant for its claim, we quantify the true structural robustness of the reasoning traces. We find that the Toulmin Coherence Score (CS) provides a consistent and automatable metric for reasoning quality, with contradiction injection producing the largest coherence degradation across both datasets. Critically, however, CS drop shows near-zero correlation with answer accuracy drop (Pearson $r < 0.08$ on both benchmarks), revealing a systematic dissociation between surface-level warrant structure and final answer correctness. This empirically confirms the *Correct by Coincidence* hypothesis: LLMs can maintain structurally coherent-sounding reasoning even as their answers fail under adversarial input conditions.

1 Introduction

Large Language Models (LLMs), as we know them today, increasingly rely on test-time computing to solve complex tasks. The rise of test-time compute - which refers to the allocation of inference-time processing via additional hidden tokens or explicit step-by-step reasoning prompts - has shifted the focus of LLM evaluation from mere output accuracy to the integrity of the intermediate thought process. Techniques such as Chain-of-Thought (CoT) prompting have demonstrated large gains in

popular benchmarks like GSM8K and StrategyQA by breaking down complex tasks into intermediate, verifiable steps.

However, the cognitive status of these reasoning traces remains contentious. Are these models truly reasoning via logical deduction, or are they performing high-dimensional pattern matching that simply mimics the guise of logic?

This project introduces a framework to audit the structural coherence of LLM reasoning under various types of adversarial noise. We argue that current evaluation metrics, such as Answer Accuracy Drop (AAD), are insufficient because they treat the model as a black box. A model might arrive at the correct answer through a "hallucinated" logical path, a phenomenon we call *Correct by Coincidence*. To address this, we leverage Argumentation Theory, specifically the Toulmin Model, to decompose reasoning traces into their functional parts: Claims, Grounds, and Warrants.

Our research addresses three critical questions:

1. **(RQ1)** To what extent can formal argumentation theory provide a granular, automated metric for LLM reasoning coherence?
2. **(RQ2)** Which specific linguistic perturbations most effectively decouple an LLM's warrant from its grounds and degrade the structural coherence of test-time compute?
3. **(RQ3)** Is there a predictive correlation between the degradation of structural warrants and the eventual failure of the final answer claim?

2 Related Work

2.1 Test-Time Compute and Reasoning Paradigms

The paradigm of reasoning has rapidly evolved around test-time compute methodologies. Wei

et al. (2022) pioneered Chain-of-Thought (CoT) prompting, demonstrating that “thinking aloud” allows models to decompose multi-step problems efficiently. More recently, the focus has shifted toward models that internalize reasoning via localized mechanisms optimized by large-scale Reinforcement Learning (RL), prompting deeper CoT behaviors.

2.2 Faithfulness and Adversarial Robustness

The “faithfulness” of LLM explanations is a growing concern in AI safety. Turpin et al. (2023) demonstrated that LLMs often provide “post-hoc rationalizations,” where the explanation does not reflect the actual cause of the prediction returned by the model. Similarly, Shi et al. (2023) showed that adding irrelevant “distractor” sentences to math problems significantly degrades performance. We extend this line of inquiry by moving beyond traditional accuracy metrics and examining the internal logical structure of the failure. We do not just ask if the model failed, but *where* in the argumentative chain the link snapped.

2.3 Argumentation Theory and LLM Evaluation

Argument Mining, an NLP process that breaks down sentences into argumentative structures, has traditionally focused on human-written text. The Toulmin Model (Toulmin, 1958) remains the gold standard for analyzing informal logic, parsing reasoning into Claims, Grounds, and Warrants. While previous works used supervised classifiers for this task, the “LLM-as-a-Judge” paradigm (Zheng et al., 2023) allows for zero-shot structural auditing. We leverage frontier models not as creative generators, but as formal logic annotators to apply this Toulmin framework to the outputs of our tested models.

3 Method and Approach

We implemented an automated evaluation pipeline that tests the resilience of an LLM’s test-time compute by injecting distracting premises into the input and auditing the resulting reasoning trace.

3.1 Dataset Preprocessing and Premise Extraction

We sampled 150 examples from the StrategyQA dataset (multi-hop implicit reasoning) and 100 examples from the GSM8K dataset (structured math word problems) using the Hugging Face datasets library.

Because StrategyQA relies heavily on unstated, multi-hop commonsense facts, it is impossible to systematically apply premise removal or contradiction perturbations without first making those hidden facts explicit. To achieve this, we deployed GPT-5.4-mini using OpenAI’s Structured Outputs (via Pydantic schema validation). For each StrategyQA example, the model acted as a formal logic annotator, ingesting the question, ground-truth answer, and provided facts, and strictly outputted the *implicit premises* required to logically deduce the answer. This successfully transformed hidden reasoning requirements into stated facts ready for perturbation.

3.2 Adversarial Perturbation Generation

We defined three specific perturbation functions applied to the input:

- **Premise Deletion:** Removing a necessary condition or fact from the problem statement.
- **Contradiction Injection:** Inserting a sentence that directly negates a necessary premise.
- **Information Shuffling:** Reordering the presentation of facts to disrupt the logical flow.

We utilized GPT-5.4-mini to generate these perturbations in a single-pass API call per example, producing a unified JSON structure containing the original input and the three perturbed variants (yielding ~1,000 total testing conditions). An important constraint was engineered into the generation prompt to prevent the model from outputting meta-commentary (e.g., preventing phrases like “*In this version, I have removed...*”). This ensured the perturbed inputs read as natural, standalone paragraphs, forcing the downstream inference models to recognize logical gaps independently.

3.3 Model Inference

In a deviation from the original proposal’s multi-model comparative structure, we narrowed our scope to conduct an in-depth structural audit of a single frontier open-weight model: llama-3.3-70b-versatile.

Inference was handled via the Groq API. To manage rate limits within the developer tier, we created an APIKeyManager class that caught HTTP 429 Rate Limit errors, parsed the server’s retry-after delays, and automatically rotated between a pool of

our API keys. For each of the 1,000 conditions, we instructed the model to “think step by step” with a temperature of 0.0 to ensure deterministic reasoning traces. The responses were parsed to separate the reasoning trace from the final answer for downstream evaluation.

3.4 Argumentative Evaluation (Judge Pipeline)

For the evaluation phase, we operationalized the Toulmin audit. A frontier judge model is tasked with extracting the specific conclusion (Claim), the evidence drawn from the prompt (Grounds), and the logical bridge explaining why the grounds justify the claim (Warrant).

At the answer level, we computed Answer Consistency Rate (ACR) and Answer Accuracy Drop (AAD). At the reasoning level, the judge model assigned a Coherence Score based on the logical strength of the Warrant, actively checking if the inference model failed gracefully, hallucinated logic, or built a flawed warrant around an injected contradiction.

4 Results

All results are based on llama-3.3-70b-versatile evaluated across 250 examples (100 GSM8K, 150 StrategyQA) under four conditions each: original, premise deletion, contradiction injection, and shuffled. Table 1 reports answer-level metrics and Table 2 reports reasoning-level coherence scores.

Dataset	Condition	Acc.	AAD	ACR
GSM8K	Original	0.950	—	—
	Premise Deletion	—	0.820	0.030
	Contradiction Inj.	—	0.690	0.000
	Shuffled	—	0.050	0.000
StrategyQA	Original	0.293	—	—
	Premise Deletion	—	-0.267	0.000
	Contradiction Inj.	—	-0.167	0.000
	Shuffled	—	-0.507	0.000

Table 1: Answer-level metrics. AAD = accuracy (original) – accuracy (perturbed). ACR = proportion of perturbed answers matching the original answer string. Negative AAD indicates higher perturbed accuracy than original.

Figure 1 shows mean CS across all four conditions by dataset. Figure 2 shows CS drop broken down by perturbation type. Figure 3 plots CS drop against accuracy drop per example with Pearson correlation fits. Figure 4 shows the proportion of

Dataset	Condition	Mean CS	CS Drop
GSM8K	Original	0.966	—
	Premise Deletion	0.960	0.005
	Contradiction Inj.	0.941	0.025
	Shuffled	0.977	-0.010
StrategyQA	Original	0.954	—
	Premise Deletion	0.962	-0.008
	Contradiction Inj.	0.936	0.017
	Shuffled	0.964	-0.011

Table 2: Reasoning-level Toulmin Coherence Scores. $CS \in [0, 1]$; higher is better. $CS \text{ Drop} = CS(\text{original}) - CS(\text{perturbed})$.

traces scoring below the Compromised threshold ($CS < 0.5$) by condition.

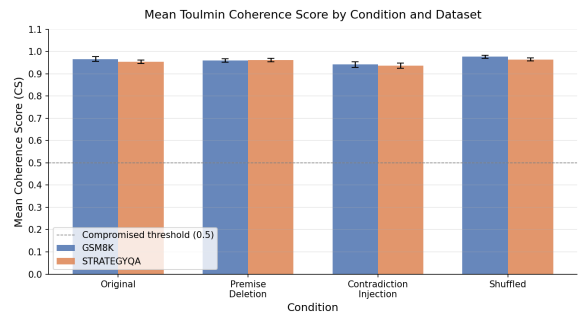


Figure 1: Mean Toulmin Coherence Score by condition and dataset. Error bars show standard error. The dashed line marks the Compromised threshold ($CS = 0.5$).

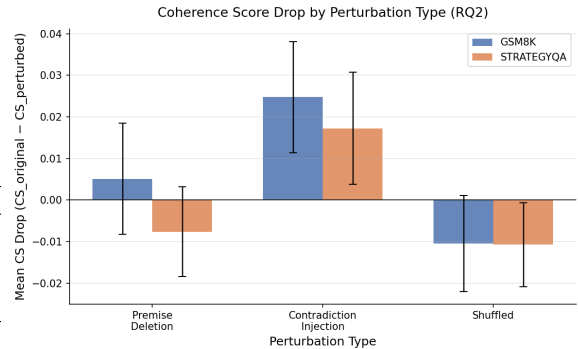


Figure 2: Mean CS drop ($CS_{\text{original}} - CS_{\text{perturbed}}$) by perturbation type and dataset. Positive values indicate coherence degradation; negative values indicate the perturbed trace was slightly more coherent.

5 Analysis and Discussion

5.1 RQ1: Toulmin CS as an Automated Coherence Metric

The Toulmin Coherence Score proves to be a workable automated metric. Across 1,000 evaluated conditions, the GPT-4o judge reliably produced

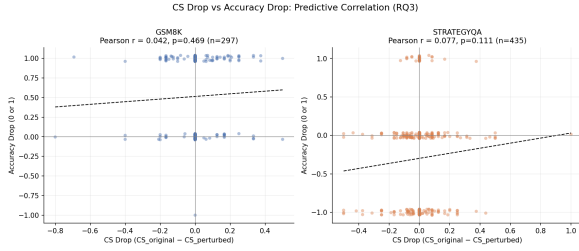


Figure 3: Scatter plot of CS drop vs. accuracy drop per perturbed example. Dashed lines show linear regression fits. Pearson $r = 0.042$ ($p = 0.469$) for GSM8K and $r = 0.077$ ($p = 0.111$) for StrategyQA.

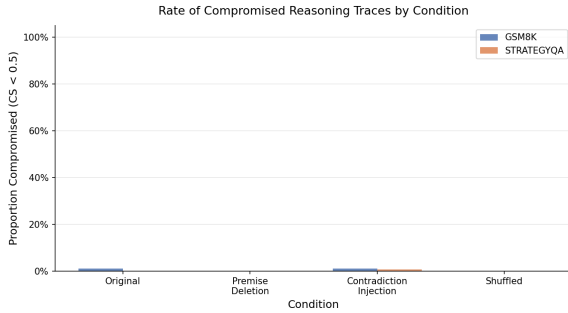


Figure 4: Proportion of reasoning traces classified as Compromised ($CS < 0.5$) by condition and dataset.

structured (Claim, Grounds, Warrant) triples and assigned warrant scores that showed meaningful variance across examples. Original CS scores ranged from 0.200 to 1.000 on GSM8K and 0.500 to 1.000 on StrategyQA, demonstrating that the metric is not degenerate. The consistency of the contradiction injection ranking across both datasets (highest CS drop in both cases) further validates that the metric is capturing something real about logical structure rather than random noise.

That said, the overall score distribution skews high, with the majority of warrant scores at 1.0. This reflects both the genuine capability of llama-3.3-70b-versatile at producing locally coherent reasoning steps and a degree of leniency in the judge, which we discuss in the Limitations section.

5.2 RQ2: Which Perturbation Degrades Coherence Most?

Contradiction injection consistently produces the largest coherence degradation across both datasets (CS drop of 0.025 on GSM8K and 0.017 on StrategyQA). This is intuitive: when a contradictory statement is injected into the input, a model that incorporates it must construct a warrant that bridges genuinely inconsistent grounds, which the judge

correctly penalizes.

Premise deletion produces only a marginal CS drop (0.005 on GSM8K, slightly negative on StrategyQA), suggesting the model either detects the missing information and flags the gap explicitly, or simply proceeds without it and generates locally valid-sounding steps anyway. Shuffling has essentially no effect on coherence in either direction, which makes sense since reordering facts does not introduce logical inconsistency — it only disrupts the flow.

5.3 RQ3: Does Structural Coherence Degradation Predict Answer Failure?

The central finding of this work is a near-zero correlation between CS drop and accuracy drop on both datasets: Pearson $r = 0.042$ ($p = 0.469$) on GSM8K and $r = 0.077$ ($p = 0.111$) on StrategyQA. Neither result approaches statistical significance.

This is most starkly visible in the GSM8K premise deletion condition, where the model’s answer accuracy drops by 0.82 — the largest collapse observed across all conditions — while the average CS drop is only 0.005. The model’s warrant structure is virtually unchanged even as its answers become overwhelmingly wrong. This is empirical evidence for the *Correct by Coincidence* phenomenon in reverse: here, the model is producing logically coherent-sounding steps that lead to incorrect conclusions. The reasoning trace sounds valid, but the underlying inference has failed because a necessary input fact was silently removed.

5.4 The StrategyQA Negative AAD

An unexpected finding is that the model achieves *higher* answer accuracy on all three perturbed conditions than on the bare original StrategyQA questions. This is explained by a design interaction: the original StrategyQA inputs are single-sentence questions requiring multi-hop implicit common-sense knowledge. The perturbed variants, however, are full paragraphs containing the explicit premises and facts needed to answer the question (with one fact deleted, contradicted, or reordered). Even a degraded paragraph containing most of the relevant context is substantially easier for the model than a bare question with no context at all. This finding highlights an important tension in perturbation-based evaluation: adding any structured context can inadvertently help rather than hurt a model that lacks the implicit knowledge required by the

original task.

6 Limitations

Single model. Due to API rate limiting during the inference stage, both datasets were evaluated using only llama-3.3-70b-versatile. The originally planned comparison between a native reasoning model (DeepSeek-R1) and a prompted CoT model (LLaMA-3.1-8B) was not completed. As a result, RQ1’s claims about model generality are limited, and any model-level comparison in the discussion is speculative.

Judge leniency and ceiling effect. Approximately 92% of individual warrant scores are 1.0, which compresses the CS distribution and limits the metric’s discriminability. This is partly attributable to the nature of the task (arithmetic steps in GSM8K have trivially valid warrants) and partly to GPT-4o scoring individual steps in isolation rather than in the context of the full flawed argument. A stricter judge prompt or a context-aware scoring approach may produce more variance.

Perturbation design for StrategyQA. As discussed above, the StrategyQA perturbation design inadvertently provides more input context than the original task, making perturbed conditions easier rather than harder. Future work should evaluate perturbations in a context-matched setting where both original and perturbed inputs provide the same amount of background information.

7 Conclusion

This work introduced an automated pipeline for auditing the structural coherence of LLM reasoning traces using the Toulmin Model of Argumentation. By applying a two-stage GPT-4o judge to decompose and score reasoning steps as (Claim, Grounds, Warrant) triples, we produced a Coherence Score metric that is measurable, consistent, and sensitive to specific classes of adversarial input perturbation.

Our central finding is that structural warrant coherence does not predict final answer correctness. A model can maintain surface-level argumentative structure while failing completely at the answer level — and conversely, can produce logically sound-sounding reasoning steps built on flawed or missing premises. This dissociation suggests that the step-by-step reasoning produced by current LLMs under Chain-of-Thought prompting reflects local syntactic coherence more than genuine inferential grounding.

Of the three perturbation types tested, contradiction injection most reliably degrades both coherence and accuracy, while shuffling has minimal effect on either. Premise deletion, despite causing the largest answer accuracy collapse on GSM8K, leaves warrant structure nearly intact — the clearest demonstration of the *Correct by Coincidence* failure mode this work set out to characterize.

Future work should extend this evaluation to native reasoning models such as DeepSeek-R1, where reasoning traces are generated through reinforcement learning rather than prompting, and to settings where the judge is prompted to evaluate warrants in the context of the full argument rather than step-by-step in isolation.

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